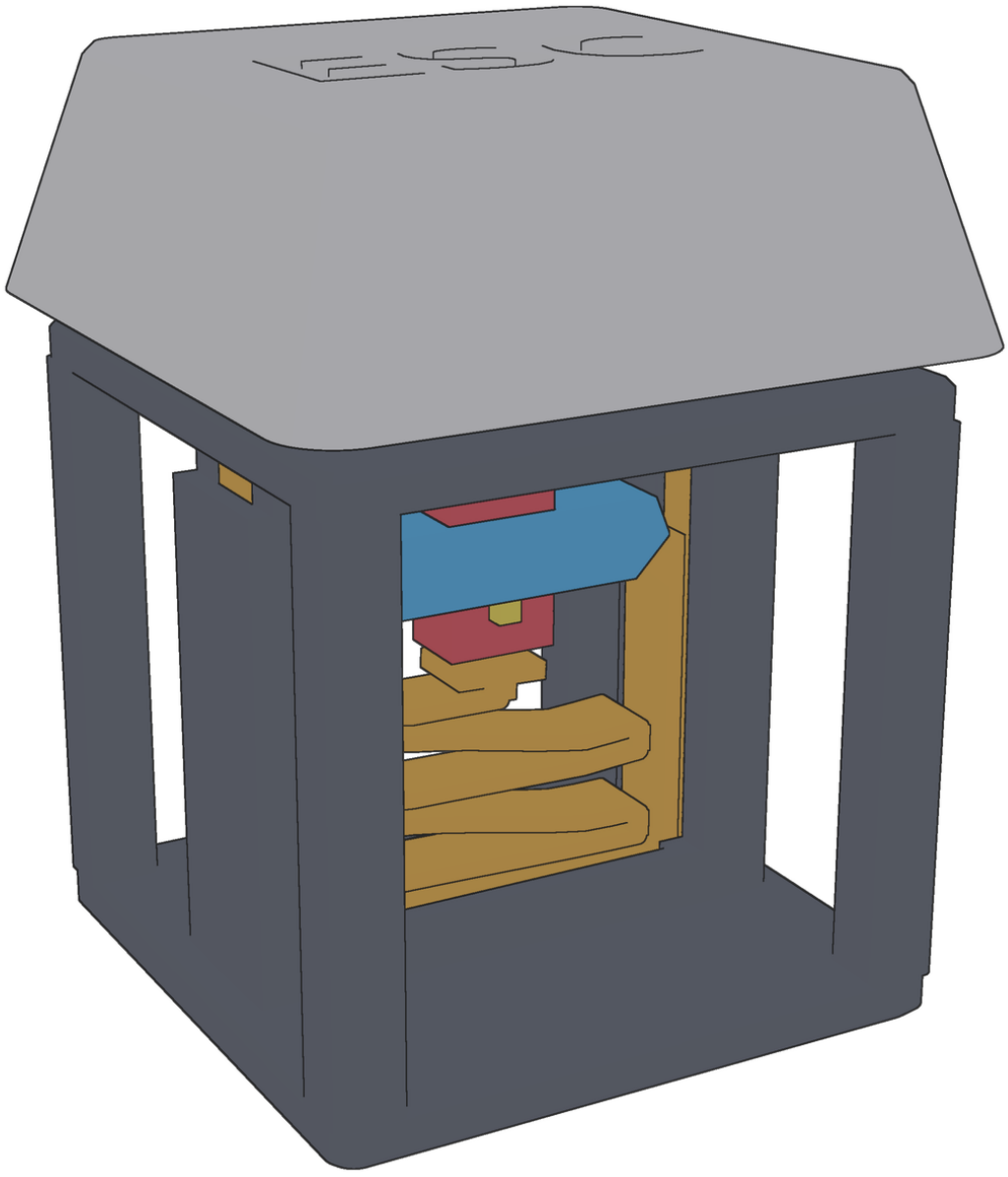
BIG KEY ESC

A giant mechanical key that really clicks. Snap-together kit, one print.

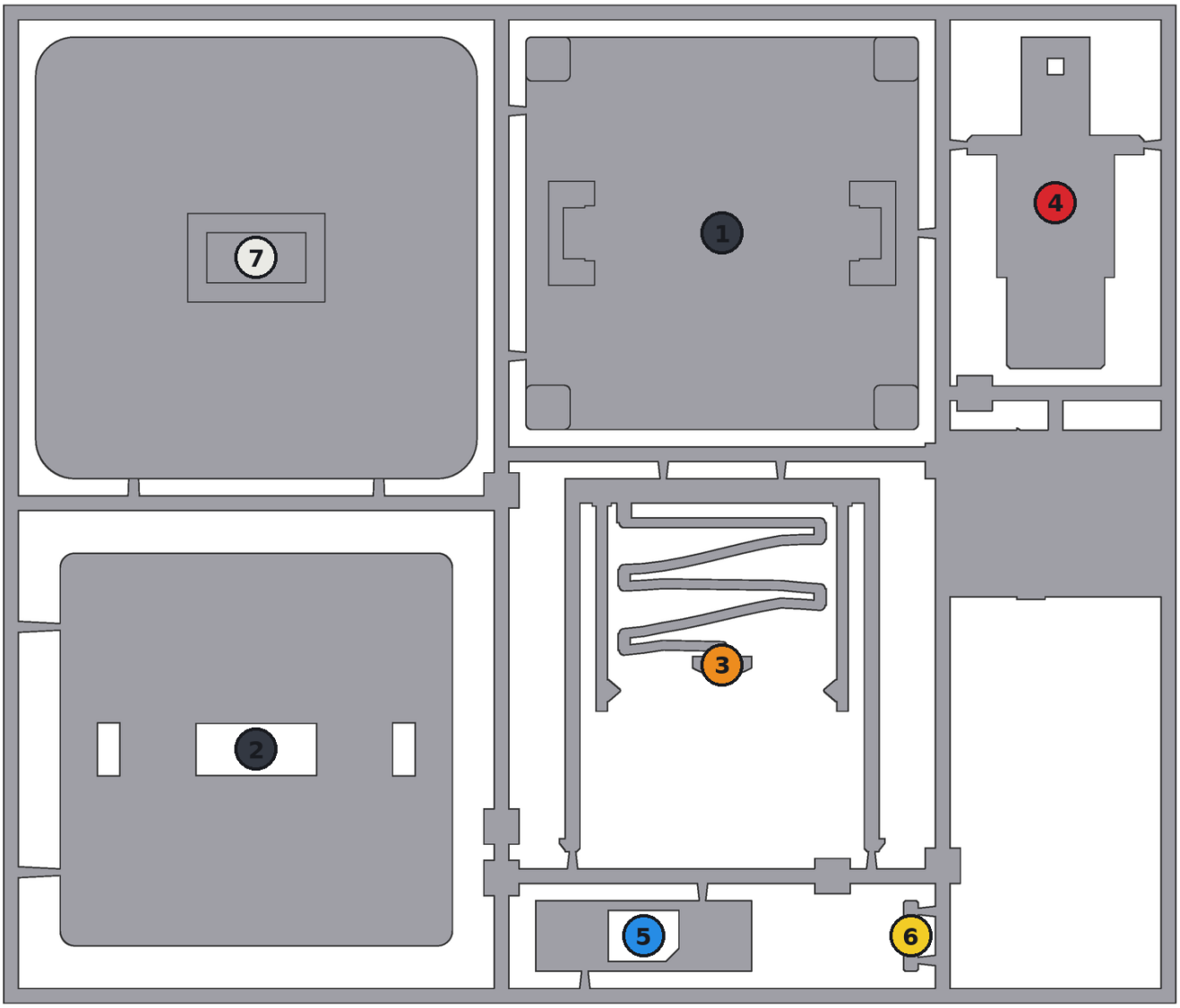


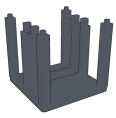
7	1	0	0	90 mm
parts	sprue	supports	glue or screws	keycap

Assembled size 80 × 80 mm base, 113 mm tall · key travel 12 mm · click at about 8 mm
Printed on a Bambu Lab P1S · 0.4 mm nozzle · PLA

What's on the sprue

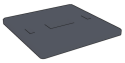
Every part is numbered on the frame. Leave parts on the sprue until a step asks for them.





1 Base

Holds everything; the open windows show the mechanism.



2 Top housing

Top housing. Guides the stem and stops it at the top.



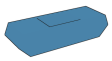
3 Spring frame

Spring frame: the zig-zag return spring, two click arms and two snap hooks in one piece.



4 Stem

Carries the keycap and slides through the top housing.



5 Click jacket

Click jacket. Snaps past the click arms and hits the pin: that is the click.



6 Retaining pin

Retaining pin. Holds the jacket on the stem.



7 Keycap

Keycap with the ESC legend.

Before you start

Print it

- Bambu Lab P1S, 0.4 mm nozzle, PLA, Textured PEI plate.
- Layer height 0.20 mm. Supports off. Brim off: the sprue frame holds every part down.
- Walls 3, top and bottom 5 layers, infill 15 % (gyroid). These settings make the springs fully solid.
- Use about 190 g of filament. Print the whole sprue in one go, in one colour.
- Tip: rub dark paint or a fine marker into the engraved ESC and wipe the top clean for a crisp legend.

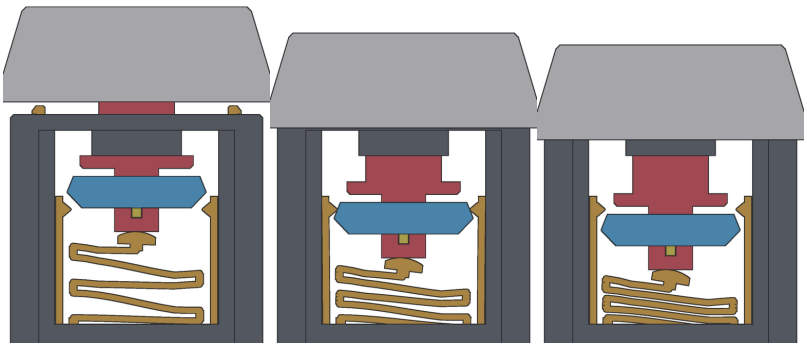
You need

- Flush cutters (sprue nippers).
- A hobby knife or a small file to clean the gate marks.

Good to know

- Cut each part off at the runner first, then trim the small stub flush with the part. The gates sit on hidden faces; on the keycap they are on the back edge.
- Clean the stubs well on the stem and the click jacket. Both slide, and a stub can make them stick.
- Every part fits only the right way. If something will not go in, turn it, don't force it.
- From step 4 until the top housing is on (step 5), keep the base standing upright.
- No glue is needed anywhere. The key can be taken apart again: squeeze the two hooks inward to lift the top housing.

How it clicks



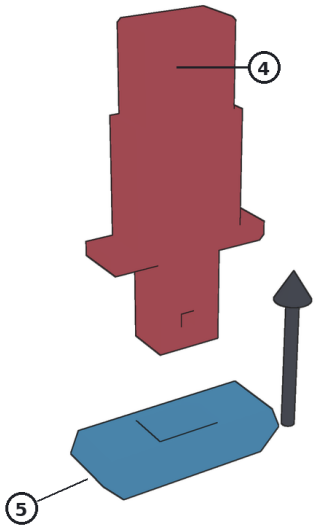
As you press, the stem pushes the blue click jacket down between two springy arms. The arms hold it back, then let go: the jacket snaps down and hits the yellow pin with a sharp click, just like a clicky keyboard switch. The orange zig-zag spring pushes everything back up.

Assembly

Steps 1–2

1 Fit the click jacket

4 Stem 5 Click jacket

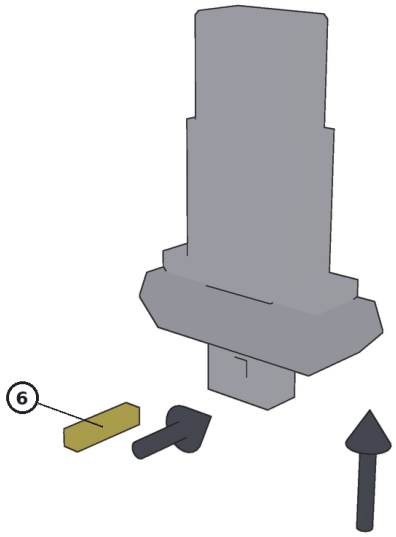


Slide the click jacket (5) up over the bottom of the stem (4), as far as it goes.

The hole has one cut corner. Line it up with the cut corner on the stem; the steep sloped side of the jacket faces down.

2 Fit the pin

6 Retaining pin



Hold the jacket up against the stem's wide shoulders. Push the pin (6) through the square hole below the jacket until it sticks out the same amount on both sides.

Let go: the jacket now rests on the pin.

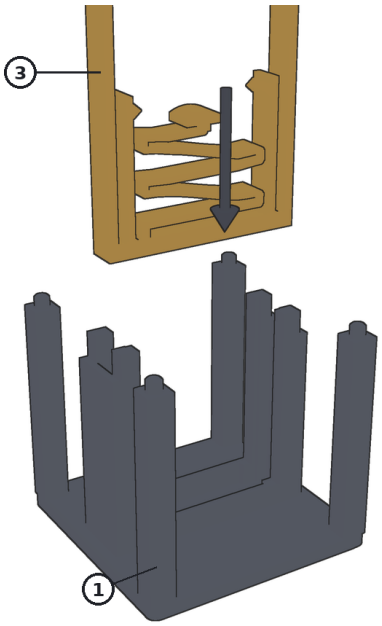
TIP
The pin slides in easily and is held by the base once the key is built. Keep the stem level until step 4.

Assembly

Steps 3–4

3 Spring frame into the base

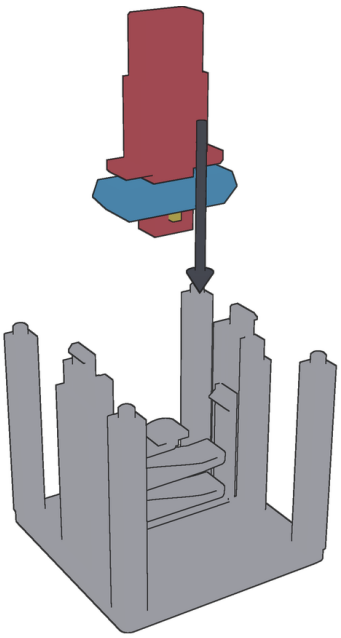
1 Base 3 Spring frame



Push the spring frame (3) straight down into the base (1), hooks at the top, until its bottom bar sits flat on the floor.

TIP
It only goes in hooks-up. The bar is a light press fit, so the frame stays put.

4 Drop in the stem



Lower the stem with its jacket into the frame, between the two click arms, until the stem rests on the round pad on top of the spring.

The jacket stops above the arms' small teeth.

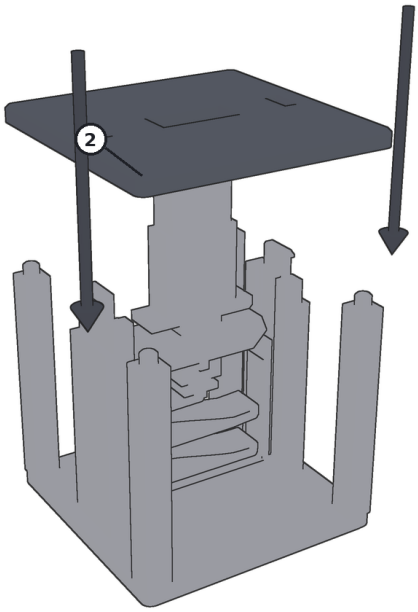
TIP
From now on keep the base upright until the top housing is on.

Assembly

Steps 5–6

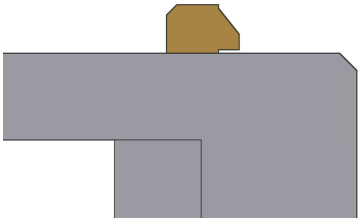
5 Close the top housing

(2) Top housing



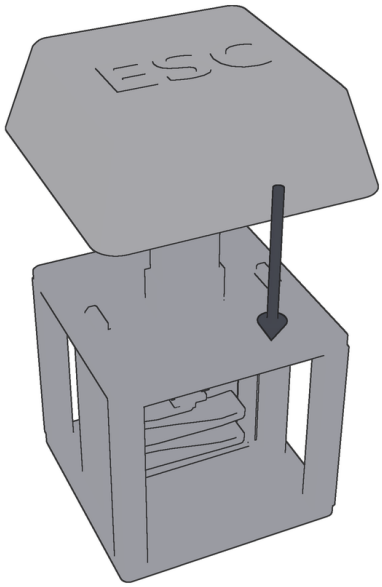
Slide the top housing (2) over the stem and press it down, compressing the spring, until both hooks click over its top and the corner pegs seat.

TIP
Check both hooks. Each one must sit on top of the housing, like this:



6 Fit the keycap

(7) Keycap

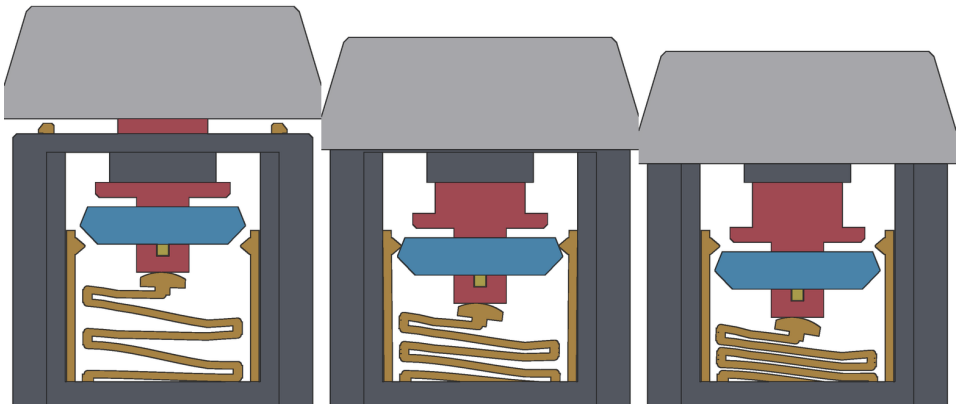


Press the keycap (7) onto the stem with ESC facing you, until it stops.

The key is finished.

Check that it works

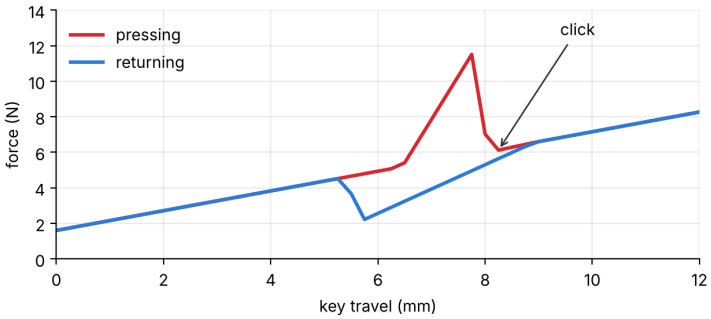
Press the middle of the keycap. At about 8 mm you feel a bump and hear a sharp click. The key bottoms out at 12 mm and springs back up, with a second, softer click on the way.



at rest **click!** **bottom**

What you will feel

Force rises gently, peaks just before the click, drops as the jacket snaps, then rises again to the bottom. Heavier than a keyboard key: press with a finger or a palm.



If something is not right

No click	Check the jacket (5) is on the stem and the pin (6) is in. Look through the window: both click arms must have their small teeth.
Key stays down or returns slowly	Trim any gate stubs on the stem (4) and jacket (5). Check both hooks are over the top housing. A little dry graphite on the stem helps.
Top housing lifts on one side	One hook is not over the edge. Press the housing down until both click.
Click too soft or too hard	Check the jacket is the right way up (steep slope down). Both click arms must have their teeth.
Keycap is loose	Push it fully home. If it still wobbles, put one layer of tape on the stem top.
Keycap will not go on	Trim the stem top lightly with a file and check the socket for a stub.
Jacket will not go on	Turn it over. It fits one way only: cut corner to cut corner, steep slope down.
Pin slides out before step 4	Normal: it is a loose fit. Hold the stem level; once inside the base the walls keep the pin in.
Base drops off when you lift the key	Add a drop of glue to each corner peg hole in the top housing.